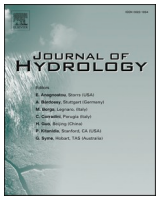
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Development of objective function-based ensemble model for streamflow forecasts

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ABSTRACT

The objective function plays an important role in hydrological model calibrations/training, since it largely determines the values of the model parameters and consequently influences the model performance. In this study, we establish two application-orientated objective functions, namely high flow balance error (HFBE) and mean squared percentage error (MSPE), for the forecasts of high flows and low flows, respectively. We examine the strengths and weaknesses of these streamflow forecast models trained with HFBE, MSPE and mean square error (MSE). Furthermore, we develop an objective function-based ensemble model (OEM) framework that can integrate the models trained with different objective functions. Our results in 273 catchments over USA show that the models trained on MSE have obvious underestimation in high-flow prediction. The models trained on HFBE can alleviate this underestimation and thus perform remarkably better for high-flow forecast. In addition, the models trained on MSPE outperform the other two models in low-flow forecast, but with an expense of the deterioration in the forecasting performance for high-flow. By incorporating the models trained on HFBE, MSPE and MSE, our proposed OEM performs well under all streamflow levels, with a median KGE of 0.96 and a median logNSE of 0.95. OEM realistically captures the mean and the variability of the observational streamflow under different scenarios with a variety of hydrometeorological conditions. This study highlights the necessity of applying objective functions that are appropriate for the modeling goal and the potential of ensemble learning methods for multi-objective optimization in hydrological modeling.

1. Introduction

Streamflow modeling and forecasting is an important topic in terms of sustainable and resilient responses to water resource management (e.g. water supply and irrigation) and extreme events prevention (e.g. extreme flood and drought) (Panagoulia and Dimou, 1997; Zhang et al., 2016). The history of rainfall–runoff modeling can date back to 170 years ago when Mulvaney applied regression-type approaches to predict the streamflow as a function of precipitation (Beven, 2012; Mulvaney, 1850). Since then, modeling approaches have been further developed by progressively understanding and encoding the physical processes through various observations and experiments as well as mathematical formulations (Black, 1996; Konapala et al., 2020). Process-based hydrologic models have been the popular choice to reasonably and explicitly model streamflow (Fatichi et al., 2016; Hipsey et al., 2015), such as the VIC (Liang et al., 1994), HBV (Bergström and Forsman, 1973), and NAM models (Nielsen and Hansen, 1973). However, due to insufficient scientific understanding and imperfect mathematical representations of hydrological processes, numerous constraints exist in process-based models (Clark et al., 2015; Vaché and McDonnell, 2006; Wagener and Gupta, 2005). In recent years, with the rapid increase of hydrometeorological data and the advances in computational power (Sun et al., 2023), machine learning, especially deep learning, has been increasingly applied in streamflow predictions (Feng et al., 2020; Gauch et al., 2021; Kratzert et al., 2021; Lin et al., 2021; Xiang et al., 2020). In general, deep learning methods do not attempt to establish models that mimic such processes as runoff generation and flow routing. Instead, they learn the mapping relationships between model inputs and outputs

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directly from the training dataset. This type of approaches can avoid the degradation of prediction accuracy due to cognitive inaccuracies of hydrologists as in the case of process-based models (Feng et al., 2020; Feng et al., 2021; Frame et al., 2021; Nearing et al., 2021), although they are criticized due to the black box properties.

Objective functions are usually used to quantify the goodness-of-fit or the prediction accuracy of model simulations against observations during model calibrations/training, guiding the optimization towards the right direction in the next iteration. In this sense, the choice of the objective function can substantially affect the quality of the calibrated/trained model (Mizukami et al., 2019). One of the most widely used objective functions is the mean squared error (MSE) or the root mean squared error (RMSE, a transformation of MSE), which is determined by the difference between simulations and the corresponding observations and the number of pairs (Sun et al., 2022). However, Armstrong and Collopy (1992) show that MSE is sensitive to sample outliers, which limits its effectiveness for calibrations. The Nash–Sutcliffe efficiency (NSE) is a normalized version of MSE (Nash and Sutcliffe, 1970), which is essentially the ratio of the forecast error to the error obtained with a naïve forecasting method (take the observational mean as forecast). And there have been numerous studies applying the NSE for model evaluations. However, Gupta et al. (2009) demonstrated that the variability tends to be underestimated in the model outcomes optimized by the NSE, leading to an overestimation in low flows and an underestimation in high flows. To mitigate this problem, Gupta et al. (2009) reformulated the Euclidean distance of the three components (variability error, mean error, and correlation between observations and predictions) decomposed from the NSE and proposed a new performance metric, namely Kling–Gupta efficiency (KGE). However, the issue of high sampling variability due to data skewness persists in the KGE, as in the NSE (Clark et al., 2021; Lamontagne et al., 2020). Although the use of the KGE introduces new problems (Liu, 2020; Santos et al., 2018), it has been gaining popularity for model calibrations and evaluations in the recent studies of hydrologic simulations (Fowler et al., 2018; Mizukami et al., 2019).

No single metric can capture all aspects of the streamflow prediction performance, and no overall best performance measures exist for model calibrations (Madsen, 2000). For example, the NSE is more sensitive to large values (Legates and McCabe Jr, 1999) and tends to ignore possible biases during dry periods (Krause et al., 2005), whereas the mean absolute percentage error (MAPE) is more affected by small values. Also, a signature on the water balance in the basin cannot represent the property of the streamflow recession curve, and vice versa, since one hydrologic signature can only reflect its corresponding streamflow characteristic.

Some studies attempt to consider as many aspects of streamflow characteristics as possible by using multiple metrics for model calibration through multi-objective optimization (Gupta et al., 1998; Gupta et al., 2008). However, it is usually difficult to reconcile the tradeoff among the optimizations for all the metrics (Mizukami et al., 2019). This may lead to the issue that no aspect of the model performance considered in the objective functions are optimal by the end of the calibration/training process (Lee and Choi, 2022), and the extent of tradeoff depends largely on the weight assigned to each optimization objective. If excessive focus is directed towards a single aspect, it may result in a detriment to the model's performance concerning other aspects.

In general, a greater emphasis is warranted on high flows if we make predictions for flood prevention, while low flows are the focus when our task is to maintain a minimum restricted water level for the safety of navigation or to maintain the ecological flow of a river (Smith et al., 2022). MSE and NSE are two commonly used model calibration/training metrics. On one hand, these two metrics are sensitive to errors in high flows while less sensitive to errors in low flows, as evidenced by the first derivative of their mathematical expressions. However, on the other hand, some studies have shown that they tend to underestimate high flows (Gupta et al., 2009; Mizukami et al., 2019), which is extremely detrimental to flood prevention. In addition, since the characteristics of high flows and low flows are generally different from each other, it is difficult to simulate them well concurrently. To addres this issue, Duan et al. (2007) and Sahraei et al. (2020) adopted different objective functions to calibrate the hydrological models, which generated the ensemble consisting of models characterized by varying objective functions. By estimating the weight to each model, they showed that ensemble-based prediction was better than the prediction from the individual model. The weights are determined by optimizing the metrics derived from simulations and observations in the historical period in their studies. This approach, while insightful, navigates us back to the intricate quandary of selecting the metric for optimization. This selection conundrum stems from the fact that various metrics exhibit dissimilar sensitivities to errors across diverse streamflow levels. Evidently, the study by Sahraei et al. (2020) exemplifies this issue. In their experiments, the optimization of weights was attained by maximizing the NSE metric between the ensemble-based simulations and measured streamflow data. Since NSE evinces limited sensitivity to errors in instances of low-flow conditions, this preference for NSE optimization inadvertently led to the ensemble-based simulations yielding suboptimal performance during arid years. Moreover, it is crucial to underscore that in the studies conducted by Duan et al. (2007) and Sahraei et al. (2020), the presupposed acquaintance with streamflow levels is a premise for engendering streamflow prediction. Nevertheless, the pragmatic translation of these methodologies into operational contexts must be tempered by the recognition of an inherent constraint: the implausibility of attaining such prescience regarding streamflow levels. Similar to Duan et al. (2007) and Sahraei et al. (2020), Chlumsky et al. (2021) incorporated weighted predictions based on various assumptions about the structure of specific processes. However, they solely employed NSE as the objective function, suggesting a bias towards fitting high-flow during the calibration period.

Ensemble learning can be regarded as, to some extent, the strategic integration of multiple models that differ in establishment method or process. There have been several ensemble strategies successfully applied in various disciplines, including hydrology and water resources. Notable among these are bagging (Breiman, 2001; Li et al., 2016), boosting (Abedi et al., 2022; Wu et al., 2020), stacking (Syarif et al., 2012; Yao et al., 2022), and blending (Van, 2012; Wu et al., 2021). In addition to the above-mentioned universal approaches, problem-orientated ensemble methods have also been proposed for particular applications, such as Gelete et al. (2023), Nourani et al. (2019), and Zhang et al. (2021). Nourani et al. (2023) stated that Ensemble models typically result in better performance than each single ensemble member, which is also shown in various hydrologic studies (Clark et al., 2008; Feng et al., 2021; Rahmani et al., 2021).

In this study, we establish two novel application-orientated objective functions for high-flow and low-flow forecasting, respectively. We verify the strength of these two application-orientated objective functions, and investigate to what degree the use of these objective functions reduces the model performance in terms of other aspects. Furthermore, we innovatively propose an objective function-based ensemble framework for streamflow predictions. In this framework, multiple models trained with various kinds of objective functions are incorporated through the ensemble learning method. In this way, we can achieve multiple goals while avoiding the tradeoff predicament existing in the model trained with a multi-objective function and therefore obtain streamflow predictions with a better overall performance. Importantly, this framework obviates the requisite dependence on antecedent knowledge pertaining to forthcoming hydrological circumstances. This distinctive attribute accentuates the framework's practical applicability, rendering it amenable for seamless integration into authentic streamflow forecasting endeavors. Moreover, we test the performance of five different algorithms as integrators in this ensemble framework. The remainder of this paper is organized as follows. Section 2 describes the methods and data. Section 3 then presents the results followed by a discussion in section 4.

Conclusions are provided in section 5.

2. Methods and data

2.1. Objective function-based ensemble model framework

Fig. 1 illustrates the implementation processes of the objective function-based ensemble model (OEM) in the courses of the model calibration/training and forecast. During the calibration/training, multiple models, called base models, are trained using the objective functions with different emphases. These base models can be process-based models or machine learning models or both. By using the specific objective function, the individual model can be applied to represent the specific aspect of streamflow, resulting in the particular strength of the model. Subsequently, during the forecasting phase, an integrator consolidates the outputs from these trained base models and generates the ultimate prediction. An integrator can be an additional model or a predefined rule, carefully designed to incorporate the base models based on their respective strengths. In this way, the OEM would achieve a multifaceted prediction.

In this study, we train three LSTM models using different objective functions as three base models. The three objective functions are designed for high flow ($Q_{0.9\text{-}1}$ bin), medium flow ($Q_{0.5\text{-}0.9}$ bin), and low flow ($Q_{0\text{-}0.5}$ bin), respectively. The numbers in these subscripts represent percentiles of streamflow series re-arranged with the ascending order. In addition, a three-category classification model is established as the integrator by using the random forest algorithm. Here, the function of this integrator is to forecast the streamflow level to determine which base model is used for generating the ultimate prediction. It should be noted that the classification model is indispensable in OEM, because it is impossible for us to know the future streamflow level in advance in the practical operation. In addition, it is imperative to note that we deliberately avoid using a single regression model to directly generate final streamflow forecasts. Rather, we use a different regression model depending on whether the flows are forecast to be high, moderate, or low. This decision is based on the understanding that using a single regression model would bring back the challenge of choosing a single objective function for the optimization process. Our approach instead allows predictions to be made from one of several models optimized to different objectives functions favoring performance on high, moderate, or low flows, depending on which of these categories the flows are expected to be in.

2.2. Three objective functions for different streamflow levels

In this study, the Mean Squared Percentage Error (MSPE), Mean Square Error (MSE), and High Flow Balance Error (HFBE) are used as the objective functions for streamflow prediction models with the emphasis on low flow ($Q_{0\text{-}0.5}$ bin), medium flow ($Q_{0.5\text{-}0.9}$ bin) and high flow ($Q_{0.9\text{-}1}$ bin), respectively. These three metrics are mathematically expressed by Equations (1)- (3). To improve the convergence speed and precision of the LSTM models, the min–max normalization (Gopal Krishna Patro and Sahu, 2015) is applied to the original data before model training and prediction. The normalization and the conversion to real values are implemented through Eqs. (4)–(5).

$$MSPE = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\left|x_{s,i}^{n} - x_{o,i}^{n}\right| \times \left(x_{o,max} - x_{o,min}\right)}{x_{o,i}^{n} \times \left(x_{o,max} - x_{o,min}\right) + x_{o,min}}\right]^{2} \quad (1)$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left(x_{o,i}^{n} - x_{s,i}^{n}\right)^{2} \quad (2)$$

$$HFBE = c \times \frac{1}{N}\sum_{i=1}^{N}\left(x_{o,i}^{n} - x_{s,i}^{n}\right)^{2} + (1-c) \times \left[\frac{1}{N_{0.9}}\sum_{h=1}^{N_{0.9}}\left(x_{o,h}^{n} - x_{s,h}^{n}\right)\right]^{2} \quad (3)$$

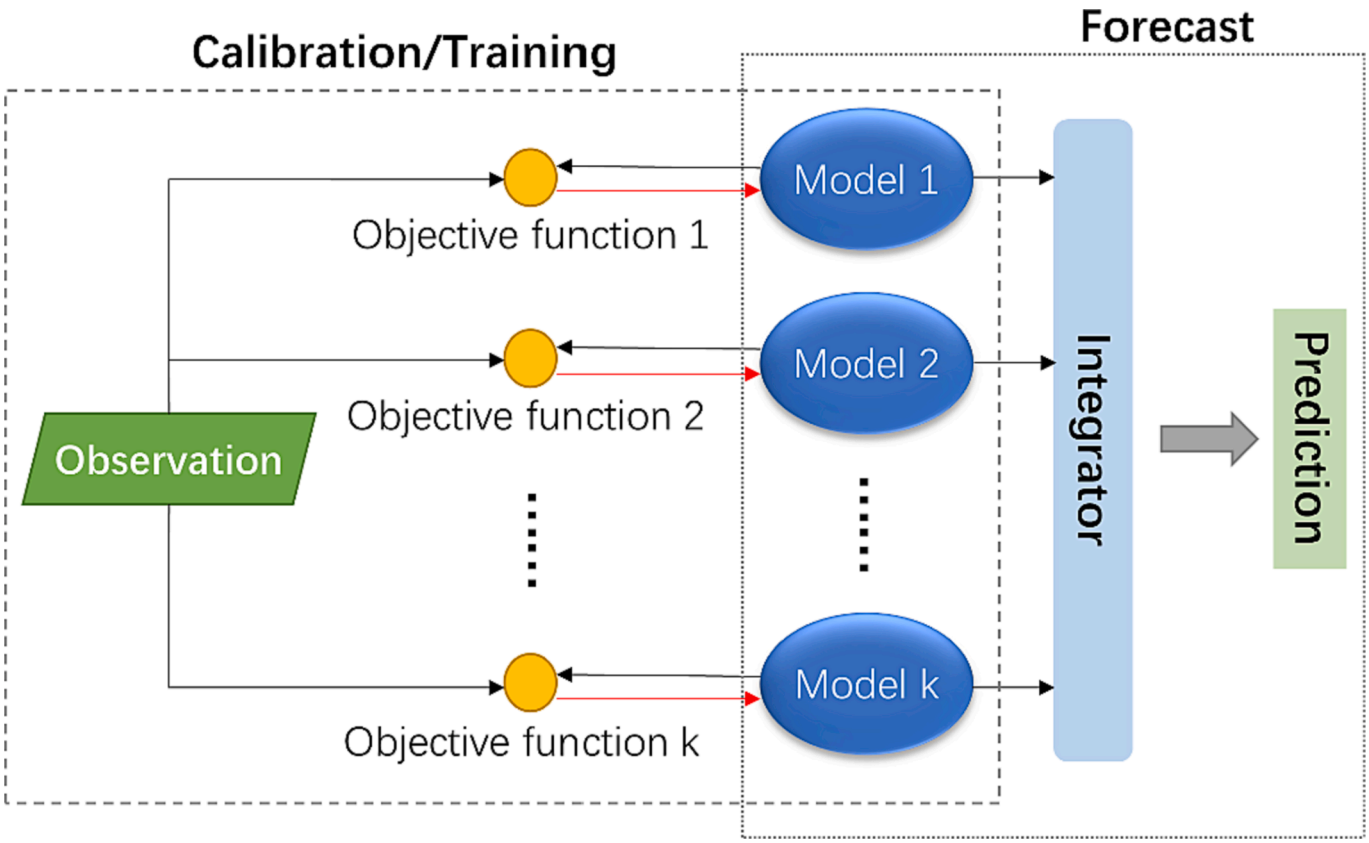


Fig. 1. The implementation processes of the objective function-based ensemble model (OEM) framework during model calibration/training and forecast.

$$x_{o,i}^{n} = \frac{x_{o,i} - x_{o,min}}{x_{o,max} - x_{o,min}} \tag{4}$$

$$x_{s,i} = x_{s,i}^{n} \times (x_{o,max} - x_{o,min}) + x_{o,min} \tag{5}$$

where $x_{o,i}$ and $x_{s,i}$ are the original values of observations and predictions, respectively, while $x_{o,i}^{n}$ and $x_{s,i}^{n}$ correspond to their respective normalized values. N is the number of all samples and $N_{0.9}$ is the number of samples in the $Q_{0.9\text{-}1}$ bin, i.e., streamflow exceeding the 90 % percentile. The c in Equation (3) is an adjustable weight coefficient between 0 and 1. $x_{o,max}$ and $x_{o,min}$ respectively represent the maximum and the minimum of the original observation.

MSE is a widely used objective function used for model training (Mizukami et al., 2019; Wang et al., 2020). However, Gupta et al. (2009) demonstrated that streamflow forecasting models calibrated with *MSE* tend to underestimate high flows while overestimating low flows. Therefore, we use *MSE* only for medium flows in this study, while we design the other two objective functions specially tailored for high and low flows, respectively. To counteract the limitations of MSE in high flow prediction, we add a penalty component that accounts for both the underestimation and overestimation of high flows. The conjunctin of the penalty with MSE results in the formulation of the *HFBE* metric. This formulation ensures a more consideration of the high flow features during the model training. The weighting factor c in *HFBE* is used to modulate the penalty degree and is set to 0.5 in this study. Conversely, MAPE demonstrates high sensitivity to errors associated with low flow forecasts. Since the value of the objective function is calculated during the model training according to the normalized observation and model output, we choose to employ *MSPE*. This metric provides better representation of the model errors with respect to the original observation, effectively addressing the gradient explosion issue inherent in MAPE. In addition, we add a squared term in the *MSPE* in order to amplify the penalty for the predictions with large mean absolute percentage errors.

2.3. KGE objective function

The variability of simulations/predictions from hydrological models is usually smaller than that of observations (Gupta et al., 2009). KGE (the equation is listed in Table 1) contains three components, i.e. correlation between observations and predictions, variability error, and mean error. Increasing the weight of the variability error term in the KGE should therefore alleviate underestimation of streamflow variability, and consequently the issue of high-flow underestimation. In light of these considerations, we experiment with three combinations of scaling factors associated with three components in this study. In the first combination, s_r, s_α and s_β are uniformly set to 1. In the second combination, s_r, s_α and s_β are adjusted to 1, 2, and 1, respectively. Lastly, the third combination assigns values of 1, 4, 1, to s_r, s_α and s_β, respectively. For convenience, KGE with these three combinations are denoted as KGE111, KGE121 and KGE141, respectively.

2.4. Algorithms used in this study

In this study, we use the Long Short-Term Memory networks (LSTM) to establish the base models for streamflow prediction. Concurrently, we adopt Random Forest (RF) to build the classification model (i.e., the integrator) for streamflow level prediction. LSTM is a special recurrent neural network that can process time series data well and has been widely used for the hydrological forecasting (Feng et al., 2020; Frame et al., 2022; Lin et al., 2023). RF is an ensemble model composed of multiple decision trees with the advantage of strong robustness. Its applicability in the field of hydrology has been examined in such studies as Shen et al. (2022) and Granata et al. (2022). To analyze the influence of the classification model on the OEM, we also incorporate four additional algorithms, namely Multilayer Perceptron (MLP), Logistics Regression (LR), K-Nearest Neighbors (KNN) and Support Vector Machine (SVM). These algorithms are applied to establish the streamflow level classification model. The predictions from these agorithms are compared with the RF-based prediction in section 3.3. Since these algorithms are not the focus of this study, we do not elaborate them here. The detailed descriptions of these algorithms can be found in the relevant literature, such as Kratzert et al. (2018) for LSTM, Shen et al. (2022) for RF, Granata et al. (2022) for MLP, Chen and Wang (2007) for LR, Dudani (1976) for KNN and Cortes and Vapnik (1995) for SVM.

For all LSTM models, the network contains two hidden layers, each of which is unidirectional and has 48 neurons. Adam (Kingma and Ba, 2014) is used to optimize the LSTM models. For the RF, MLP, LR, KNN and SVM models, the default settings inherent in the scikit-learn 1.2.0 package in Python are adopted. It should be noted that while we do not specifically optimize the hyperparameters of each individual model, their settings remain identical, thereby ensuring a fair and equitable comparison of their forecast performance. The features used in the models are described in Section 2.6.

Table 1
The evaluation metrics used in this study.

Name	Equation	Range of values
KGE (Gupta et al., 2009)	$KGE = 1 - \sqrt{s_r \bullet (r-1)^2 + s_\alpha \bullet (\alpha - 1)^2 + s_\beta \bullet (\beta - 1)^2}$ $r = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_{o,i} - \mu_s)(x_{m,i} - \mu_s)}{\sqrt{\sum_{i=1}^{N}(x_{o,i} - \mu_o)^2}\sqrt{\sum_{i=1}^{N}(x_{s,i} - \mu_s)^2}}$ $\alpha = \frac{\sigma_s}{\sigma_o}$ $\beta = \frac{\mu_s}{\mu_o}$	(-∞, 1] 1 is the best
logNSE (Nash and Sutcliffe, 1970)	$logNSE = 1 - \frac{\sum_{i=1}^{N}(log(x_{o,i}) - \log(x_{s,i}))^2}{\sum_{i=1}^{N}(log(x_{o,i}) - log(\mu_o))^2}$	(-∞, 1] 1 is the best
Pearson-r	$Pearson\text{-}r = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_{o,i} - \mu_s)(x_{m,i} - \mu_s)}{\sqrt{\sum_{i=1}^{N}(x_{o,i} - \mu_o)^2}\sqrt{\sum_{i=1}^{N}(x_{s,i} - \mu_s)^2}}$	[-1, 1] 1 is the best
Beta-KGE (Gupta et al., 2009)	$Beta\text{-}KGE = \frac{\mu_s}{\mu_o}$	[0, +∞) 1 is the best
OUEA	$OUEA = \sum_{i=1}^{N} ou_i$	[-N, N] 0 is the best
Precision	$Precision = \frac{TP}{TP + FP}$	[0, 1] 1 is the best
Recall	$Recall = \frac{TP}{TP + FN}$	[0, 1] 1 is the best
F1	$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$	[0, 1] 1 is the best

2.5. Evaluation metrics

The evaluation metrics used in this study are summarized in Table 1. Within this context, $x_{o,i}$ and $x_{s,i}$ respectively represent the observation and model prediction. N is the total number of samples. μ_o and μ_s are the mean of the observation and prediction respectively, while σ_o and σ_s are the standard deviation of the observation and prediction respectively. s_r, s_α, and s_β in the KGE are the scaling factors that can be used to re-scale the criteria space. For the metric OUEA, if $x_{s,i} - x_{o,i} > 0$, $ou_i = 1$, otherwise $ou_i = -1$. For the metrics Precision and Recall, TP, FP, and FN represent the numbers of true positives, false positives, and false negatives, respectively.

Several evaluation metrics are employed to comprehensively assess the performance of streamflow forecasting models in this study. Standard KGE (i.e. s_r, s_α, and s_β are uniformly set to 1) stands out as a widely used metric for evaluating the overall effectiveness of streamflow fore-

casting. logNSE, on the other hand, is attuned to detecting errors in low-flow predictions. Within the context of different streamflow levels, metrics such as Pearson-r, Beta-KGE, and OUEA are used to evaluate the predictive capability of models. In the categorical realm, the metrics Precision, Recall, and F1 are used to evaluate the categorical effectiveness of the integrator of the OEM for each streamflow class.

2.6. Data

The streamflow data used in this study are from the Global Runoff Data Centre (GRDC), an entity responsible for managing the river discharge data at more than 10,000 stations in 159 countries. We select 273 stations in the continental United States that have complete daily streamflow data series from 1982 to 2018. The locations of these selected stations are shown in Fig. 2. In addition, the total precipitation, 2-m above ground temperature, and total potential evaporation from the North America Land Data Assimilation System version 2 (NLDAS-2) (Xia et al., 2012) are also used as auxiliary inputs for streamflow prediction.

In this study, the inputs of LSTM (i.e., base models in OEM) and classification models (i.e., classification model in OEM) are identical. These inputs include daily streamflow, daily precipitation, daily 2-m air temperature, and daily potential evaporation. Fig. 3 shows the Pearson correlation coefficient between the streamflow and these four variables with different time lags. We can observe that when the time lag is greater than 10 days, the correlation coefficients of precipitation for all watersheds are consistently below 0.4, and the values gradually decrease with the increase of the time lag. In the case of temperature and evapotranspiration, the correlation patterns vary significantly among different watersheds. Antecedent streamflow, particularly with a 1-day time lag shows relatively high correlation across all watersheds. For the sake of standardization, the model inputs for all watersheds encompass all four variables in the past 10 days, resulting in a total of 40 inputs in one sample. It should be noted that this may not be the optimal configuration for specific watersheds, and optimizing the model configuration for each watershed is not the focus of this study.

To effectively train the models, the data from 1982 to 2013 are employed, 90 % and 10 % of which in chronological order are taken as the training and verification sets, respectively. We allocate 90 % of the data to the training set because we aim to have more data available for model training, enhancing the robustness of the models. The remaining data spanning from 2014 to 2018 are designated as the testing set. Note that all the model evaluations are based on prediction from the testing set in this study.

3. Results

In this section, for convenience, the LSTM models trained with MSE, HFBE, and MSPE are represented by $LSTM_{MSE}$, $LSTM_{HFBE,}$ and $LSTM_{MSPE}$, respectively. OEM refers to the objective function-based ensemble model with $LSTM_{MSE}$, $LSTM_{HFBE,}$ and $LSTM_{MSPE}$ as the base models and RF as the integrator. In addition, the LSTM models trained with the objective functions of KGE111, KGE121 and KGE141 are represented by $LSTM_{KGE111}$, $LSTM_{KGE121,}$ and $LSTM_{KGE141}$, respectively.

3.1. Evaluation of models trained with three different objective functions

As shown in Fig. 4a, when the entire testing data is used for evaluation, the performance of $LSTM_{MSPE}$, $LSTM_{MSE}$ and $LSTM_{HFBE}$ are comparable in term of Pearson-r. However, when the forecast performances are respectively evaluated under different streamflow levels, things are very different. $LSTM_{MSPE}$ is significantly superior to $LSTM_{MSE}$ and $LSTM_{HFBE}$ for low-flow and medium-flow forecasting, while inferior to them for high-flow forecasting.

According to the metric Beta-KGE (Fig. 4b), the prediction from $LSTM_{MSE}$ can well resemble the mean of observations when evaluated with the whole testing set. However, it tends to overestimate for low-flow forecasting while underestimating for high-flow forecasting. In addition, the predictions from $LSTM_{MSE}$ for lower flows have larger uncertainties, as indicated by the wider ranges of the box plots for $Q_{0\text{-}0.2}$ and $Q_{0.2\text{-}0.4}$. Compared with $LSTM_{MSE}$, $LSTM_{HFBE}$ effectively alleviates the underestimation for high-flow forecasting, but with an expense of a degradation in other aspects, which is in line with our expectations for $LSTM_{HFBE}$. Moreover, $LSTM_{MSPE}$ obtains a better evaluation than the other two models for low-flow forecasting ($Q_{0\text{-}0.2}$ and $Q_{0.2\text{-}0.4}$).

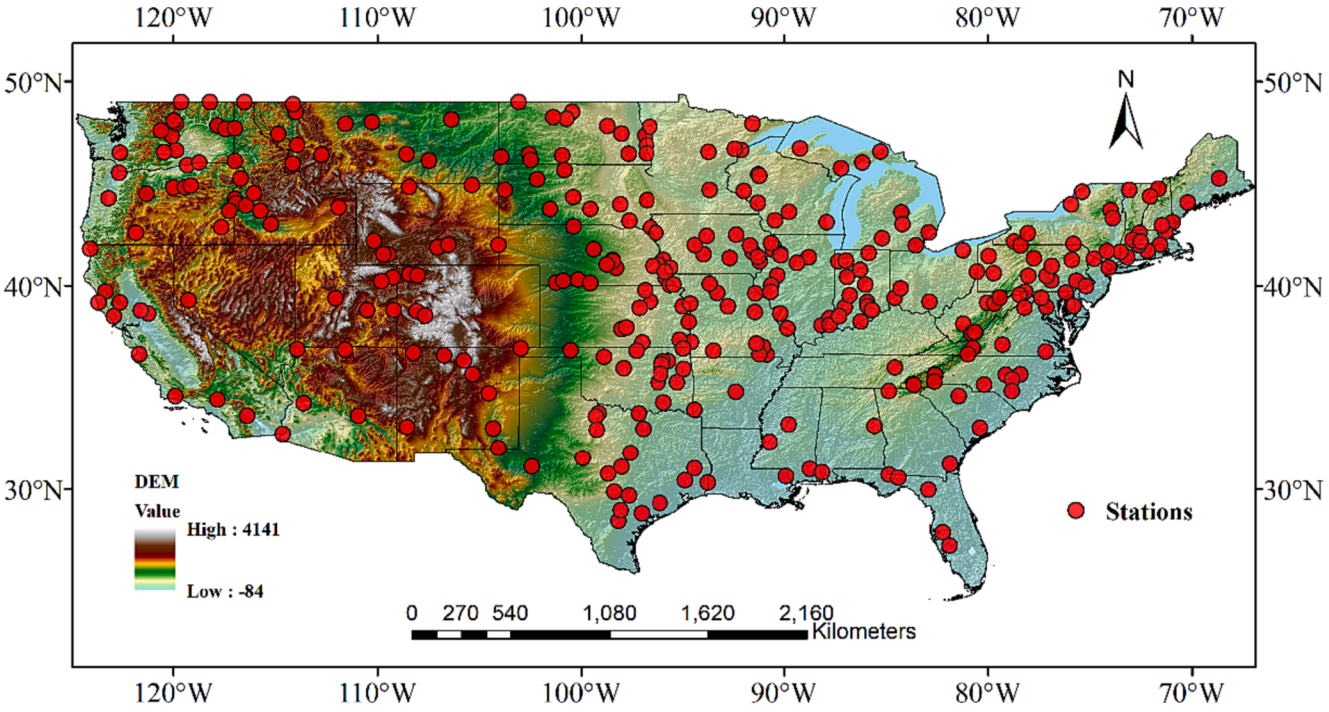


Fig. 2. The spatial distribution of hydrological stations corresponds to 273 watersheds used in this study.

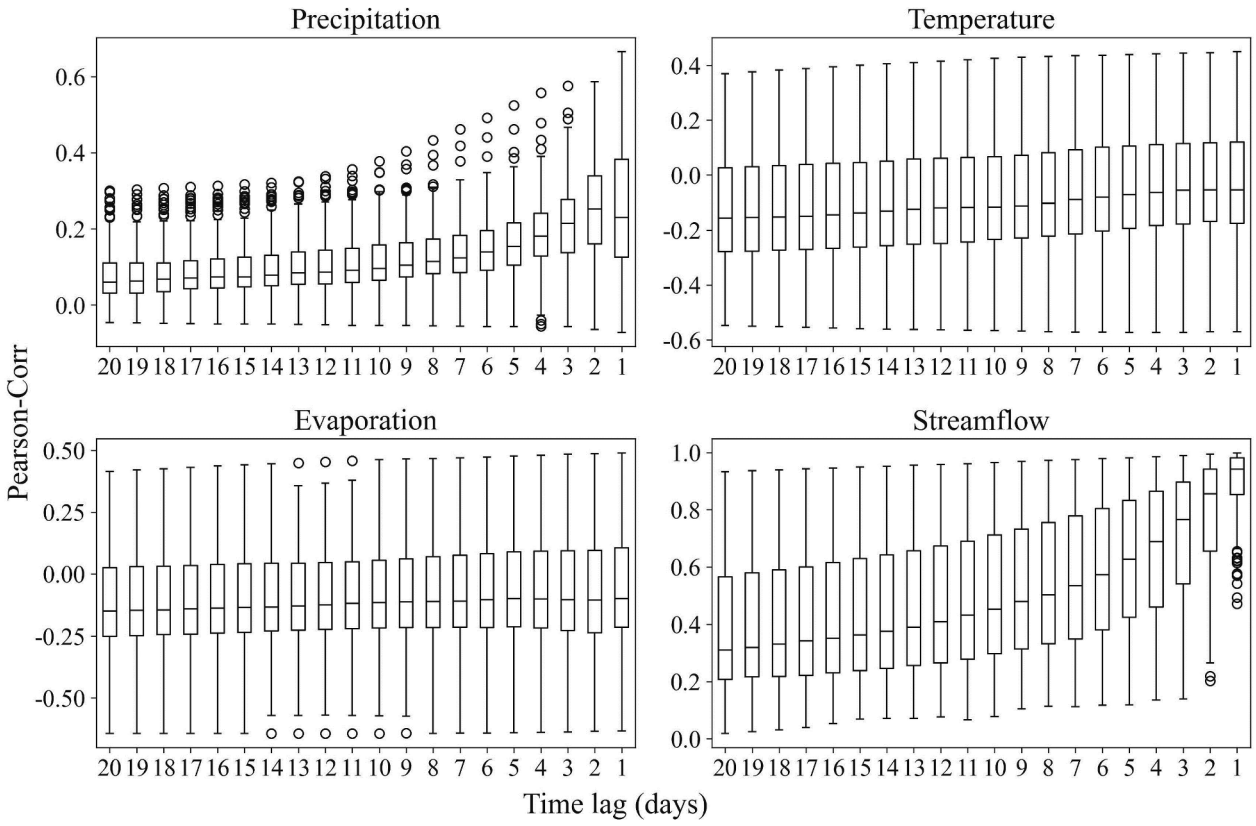


Fig. 3. The Pearson correlation coefficient between the streamflow and four variables with different time lags.

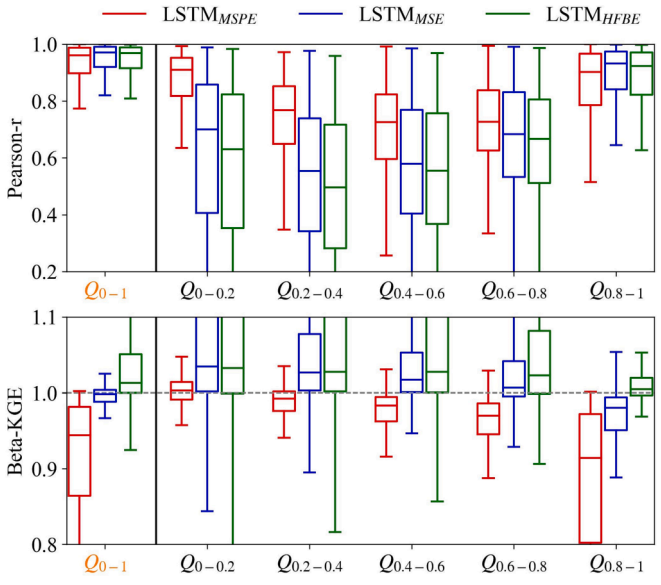


Fig. 4. The forecasting performance of $LSTM_{MSPE}$, $LSTM_{MSE}$ and $LSTM_{HFBE}$ in terms of the metrics Pearson-r and Beta-KGE over 273 watersheds. In each panel, the three box plots on the leftmost groug represent the distribution of the metric calculated using the entire testing dataset, while the other five groups to the right respectively refer to the distribution of the metric calculated based on 20% data associated with certain streamflow level from low to high. For example, the second left group of box plots is for the streamflow falling below the 20% percentile, while the third left group of box plots corresponds to the streamflow within the 20%–40% percentile range.

3.2. Evaluation of the OEM

Fig. 5a illustrates the runoff prediction performance of the OEM across 273 watersheds. Despite some watersheds exhibiting relatively poorer performance, the OEM demonstrates high KGE and logNSE values in most of the watersheds. According to Fig. 5b, the KGE of OEM is slightly higher than those of $LSTM_{KGE111}$, $LSTM_{KGE121}$, and $LSTM_{KGE141}$. At the same time, OEM greatly improves the streamflow forecast in term of metric logNSE compared with the other three models. The median logNSE for OEM is 0.95, whereas the respective median logNSE values for $LSTM_{KGE111}$, $LSTM_{KGE121}$ and $LSTM_{KGE141}$ are 0.87, 0.85 and 0.84.

Overall, OEM's predictions are more closely aligned with observations. This holds true whether conducting across the entire testing dataset or conducting separately based on varying streamflow levels. As shown in Fig. 6, the Beta-KGE of the OEM is closer to 1 than the other three models accoss all streamflow levels. It can be found that the OEM effectively alleviates the underestimation for high-flow ($Q_{0.8\text{-}1}$) forecasting. In contrast, $LSTM_{KGE121}$ and $LSTM_{KGE141}$ do not show improvement in high-flow forecasting compared with $LSTM_{KGE111}$. Instead, these two models tend to exacerbate the problem of underestimation of low-flow ($Q_{0\text{-}0.2}$) forecasts. This indicates that increasing the weight of variability in the KGE does not help the model to overcome the issue of underestimating high-flow forecasts.

To understand the superior performance of OEM, we calculate the OUEA values of OEM and its three base models for different streamflow levels. As indicated by OUEA in Fig. 7a, the predictions for very low flows from $LSTM_{MSE}$ are likely to be overestimated, and the overestimation decreases with the increase of the streamflow level. For high flows, however, $LSTM_{MSE}$ has a strong tendency to underestimate the streamflow compared with observations. According to Fig. 7b, the predictions from $LSTM_{HFBE}$ are more likely to be greater than the observations, even at a high streamflow level, which is consistent with the

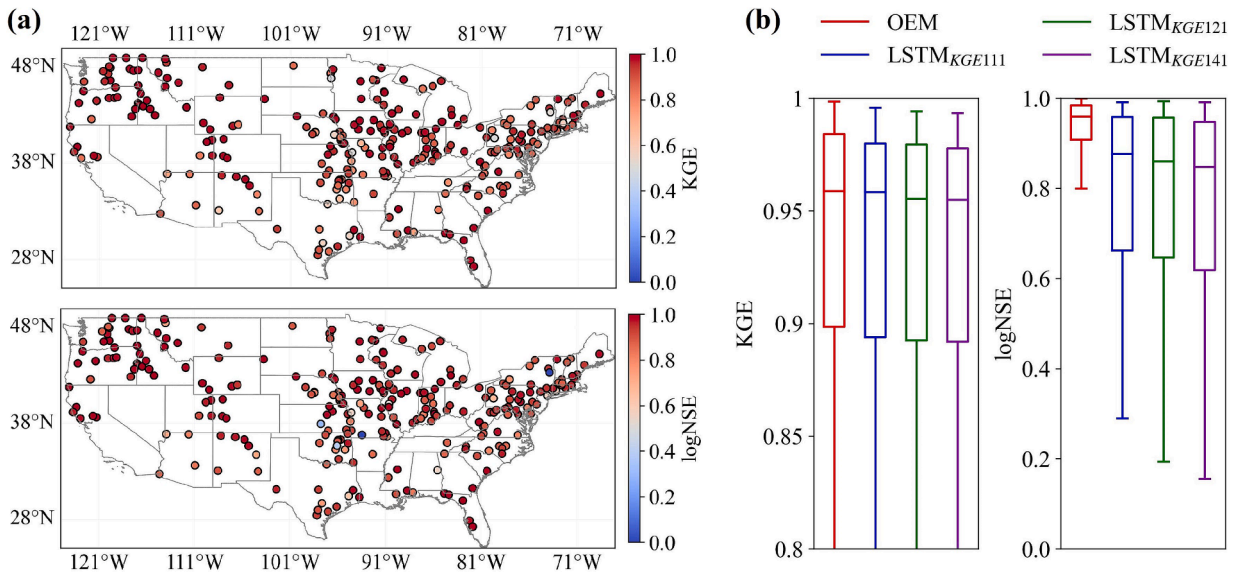


Fig. 5. (a) The forecasting performance of OEM in terms of the metrics KGE and logNSE over 273 watersheds. (b) The comparison among the forecasting performance of OEM, $LSTM_{KGE111}$, $LSTM_{KGE121}$, and $LSTM_{KGE141}$.

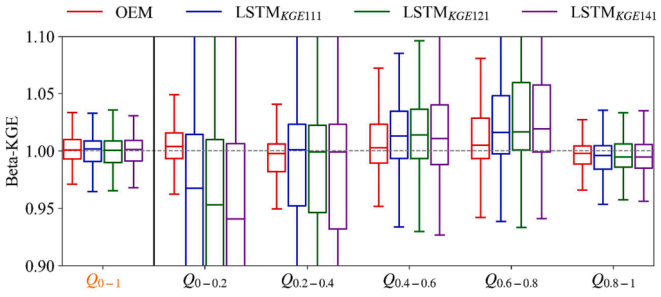


Fig. 6. The forecasting performance of OEM, $LSTM_{KGE111}$, $LSTM_{KGE121}$, and $LSTM_{KGE141}$ in terms of the metric Beta-KGE over 273 watersheds. Box plots represent the same distributions in terms of streamflow level classification as in Fig. 4.

results shown in Fig. 4. And the performance of $LSTM_{HFBE}$ for very high flows is better than that of $LSTM_{MSE}$. The OUEA of $LSTM_{MSPE}$ (Fig. 7c) is closer to zero in the low and medium flows compared with that of $LSTM_{MSE}$ and $LSTM_{HFBE}$. According to Fig. 7d, the OUEA of the OEM in all streamflow levels are smaller than those of $LSTM_{MSE}$, $LSTM_{HFBE}$, and $LSTM_{MSPE}$, which indicates that the OEM can achieve more realistic predictions in all streamflow levels by aptly integrating the three base models.

To further verify the robustness of the OEM, we conduct assessments under four different hydrometeorological scenarios, as outlined in Table 2. The four scenarios are generated by selecting samples based on the mean or the variability of the time series in a year. Specifically, for a given watershed, the scenarios are defined as follows:

1. The dry scenario corresponds to years in which the annual average streamflow in the testing set is lower than the corresponding minimum annual average value in the training set (1982–2013).
2. Similarly, the wet secenrio refers to years in which the annual average streamflow in the testing set is higher than the corresponding maximum annual average value in the training set.
3. The homogenous scenario encompasses years in which the variance of daily streamflow falls below the corresponding minimum value in the training set.
4. The heterogenous scenario includes years in which the variance is greater than the maximum in the training set.

The numbers of the watersheds under dry, wet, homogenous, and heterogeneous scenarios are 15, 39, 25, and 51, respectively. According to Table 2, the OEM can still play an effective role under all four scenarios, which is supported by the improvements of both KGE and logNSE from $LSTM_{KGE111}$ to the OEM. The improvement for the low-flow forecasting is larger in the dry and homogenous scenarios than in the wet and heterogenous scenarios.

3.3. Evaluation of the integrator of OEM

The integrator is a model component through which the base models are incorporated into the OEM. In this study, the integrator is essentially a classification model, and we adopt the RF algorithm as the integrator in the above experiments. To examine the effectiveness of the RF algorithm, we apply four different algorithms, namely MLP, LR, KNN, and SVM, to classify the streamflow level. The comparison of these five algorithms is shown in Fig. 8. Three evaluation metrics, Precision, Recall, and F1, are used here, and their equations are listed in Table 1.

It is evident from the analysis in Fig. 8 that none of the employed methodologies exhibit absolute precision in prognosticating forthcoming streamflow levels. This empirical reality is congruent with the intricacies inherent in authentic hydrological forecasting operations, where the intricate interplay of multifaceted variables invariably leads to a degree of predictive uncertainty. In general, RF is the best, with a median F1 greater than 0.9, followed by MLP and LR. While KNN and SVM are relatively worse compared with the preceding three algorithms, their medians of F1 are still greater than 0.85. The classification performance also differs with the streamflow level. The model prediction is usually better for low flows than high flows, and is in between for medium flows. The relatively low Recalls for high flows are found in all five classification algorithms, suggesting that a number of high flows are wrongly predicted as medium or low flows. This may be caused by the relatively limted proportion of high flows in the dataset.

4. Discussion

4.1. The influence of objective function on model performance

Fig. 9 illustrates the training process of a Long Short-Term Memory network (LSTM) model for streamflow forecasting, which is trained with the MSE objective function. The evolutions of the loss function as well as

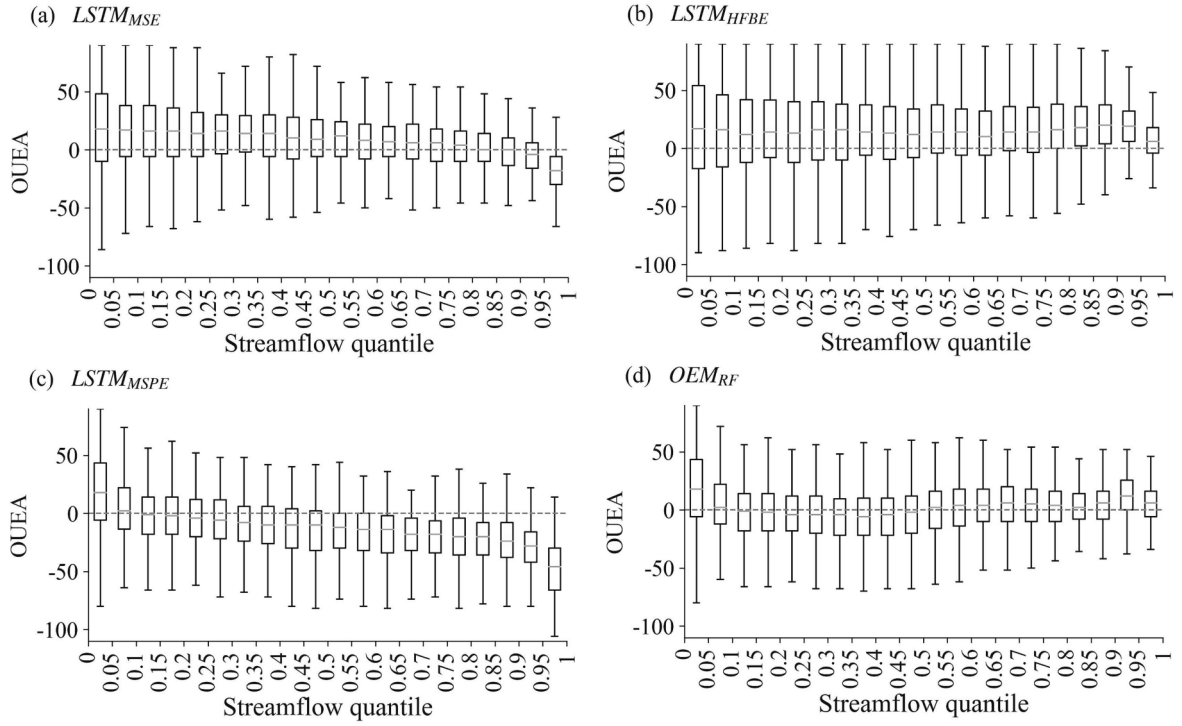


Fig. 7. The over- and under- estimate accumulation (OUEA) under different streamflow levels for (a) $LSTM_{MSE}$, (b) $LSTM_{HFBE}$, (c) $LSTM_{MSPE}$, and (d) OEM. The testing set is divided into 20 equitably sized bins, organized based on ascending streamflow levels from low to high. The OUEA is calculated individually for each bin. The box plots above the zero line indicate that there are more overestimations in the corresponding streamflow levels compared to underestimations, and those below the zero line imply more underestimations. Evidently, the base models $LSTM_{MSE}$, $LSTM_{HFBE}$, and $LSTM_{MSPE}$ are skilled in the predictions in a certain streamflow level, respectively. In contrast, the OEM performs well accross all levels by leveraging the strength of the three base models.

Table 2
The medians of KGE and logNSE for $LSTM_{KGE111}$ and the OEM under four different hydrometeorological scenarios. The numbers in brackets in the first row represent the numbers of watersheds.

		Dry (15)	Wet (39)	Homogenous (25)	Heterogenous (51)
$LSTM_{KGE111}$	KGE	0.87	0.95	0.77	0.98
	logNSE	0.34	0.93	0.19	0.97
OEM	KGE	0.91	0.96	0.80	0.98
	logNSE	0.95	0.95	0.87	0.98

five evaluation metrics with model iteration are shown for the training and validation sets, respectively. The moments when these five metrics for the validation set reach optimality are marked by black dots, together with the corresponding numbers of model iterations (N_{opt}). Due to the similarity of the mathematical formulation between RMSE and the objective function MSE, their trajectories are almost identical throughout the entire training process, including N_{opt}. However, the other four metrics diverge from the objective function in terms of both pathway and N_{opt}.

Due to the imperfectness of forecasting models and the errors with hydrometeorological data, there is always a deviation between the model simulation/forecast and the observation. The choice of the objective function greatly influence the model parameters and consequently the model robustness. Usually, the model calibrated/trained with an objective function performs better on the corresponding evaluation metric, when compared with the model calibrated/trained with a different objective function. However, the former may perform worse than the latter in terms of other metrics or aspects. In our study, the models trained with MSPE are superior to the models trained with HFBE for low flows but inferior for high flows. The models trained with HFBE outperform the models trained with MSE in terms of the consistency

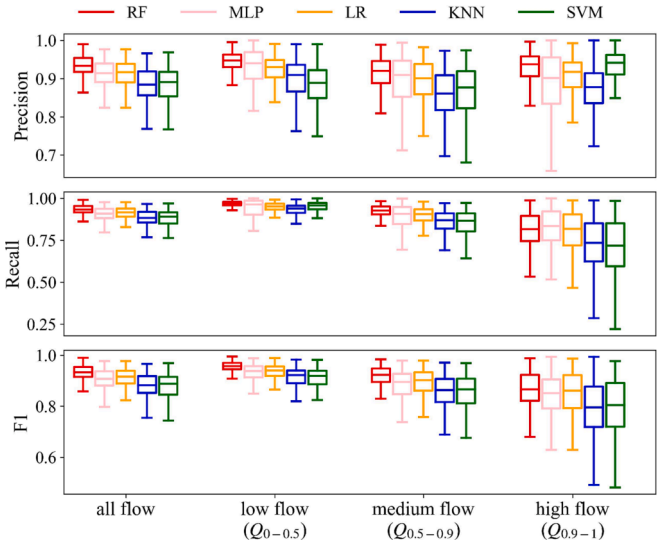


Fig. 8. The performances of RF, MLP, LR, KNN, and SVM for streamflow level classification. The box plots of three metrics (precision, recall and F1) derived from these five models are shown in different colors.

with observations for high flows, but at an expense of the deterioration in RMSE. Our results indicate that no objective function can produce the best performance in all aspects for the streamflow forecasting.

4.2. Relative importance between objective function and algorithm used for model establishment

It is a common practice to seek "better" algorithms when the model

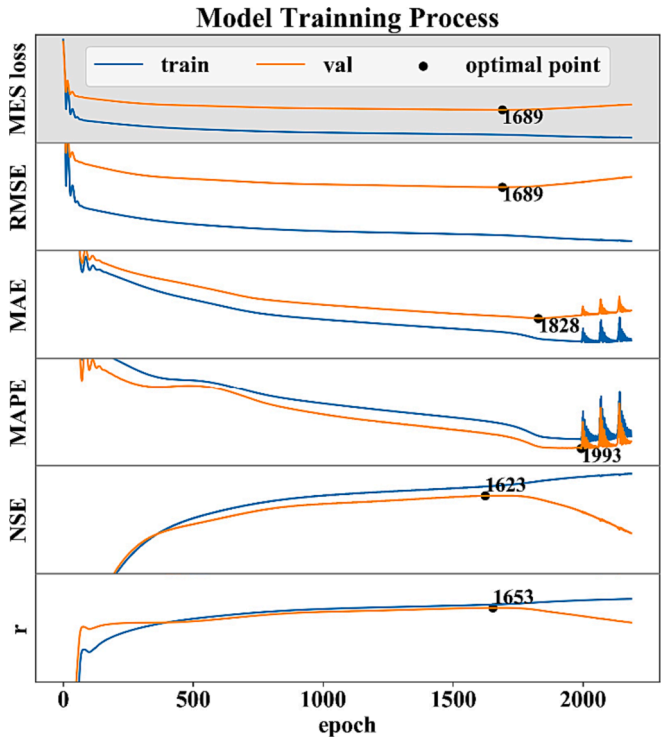


Fig. 9. The evolution of the objective function as well as five evaluation metrics during the training process of an LSTM model trained with MSE objective function for streamflow forecasting.

prediction is unsatisfactory in a particular aspect. However, selecting an appropriate objective function might be more effective in improving the model performance. To explore the relative importance between the objective function and the algorithm for the model establishment, we additionally apply the MLP to establish the streamflow forecasting models trained with MSE, HFBE, and MSPE, respectively, which are denoted by MLP_{MSE}, MLP_{HFBE}, and MLP_{MSPE}, respectively. According to Fig. 10, when both MLP and LSTM are trained with the same metric MSE, MLP_{MSE} is inferior to $LSTM_{MSE}$ in the predictions for both low flows and high flows. However, the MLP_{MSPE} is significantly superior to the $LSTM_{MSE}$ in low flow forecast, which is supported by a higher correlation coefficient and a better Beta-KGE from $LSTM_{MSE}$ to MLP_{MSPE}. For the high flow forecast, the MLP_{HFBE} considerably outperforms the $LSTM_{MSE}$ in terms of the metric Beta-KGE. The proportions of Beta-KGE for high flows in the range of 0.9–1.1 and 0.95–1.05 are increased from 83 % and 66 % for the $LSTM_{MSE}$ to 97 % and 90 % for the MLP_{HFBE}, respectively. Evidently, the objective function selected for the model training can be more important than the algorithm selected for the model establishment. For an algorithm that does not perform well, changing the objective function can lead to a considerable improvement in the model performance regarding the aspects appropriately reflected by the metric used in this objective function.

4.3. Application-orientated objective function

Verifying the soundness of a given model is a process to prove that its output is “adequately accurate”. However, “adequately accurate” usually varies with the specific objective of the project (Refsgaard, 1997). Because different projects have different focuses on the streamflow forecast, using an application-orientated objective function is of great significance, as the objective function directly reflect the project focus (Smith et al., 2022). In this study, we design the MSPE and HFBE for low-flow and high-flow forecasts, respectively. The former substantially improves the overall performance of the forecast model for low flows, while the latter helps better match the observational high flow. In the context of prioritizing maintenance of the river’s ecological flow, the MSPE would be an appropriate objective function for the model training. By contrast, HFBE would be the better choice when the purpose is to

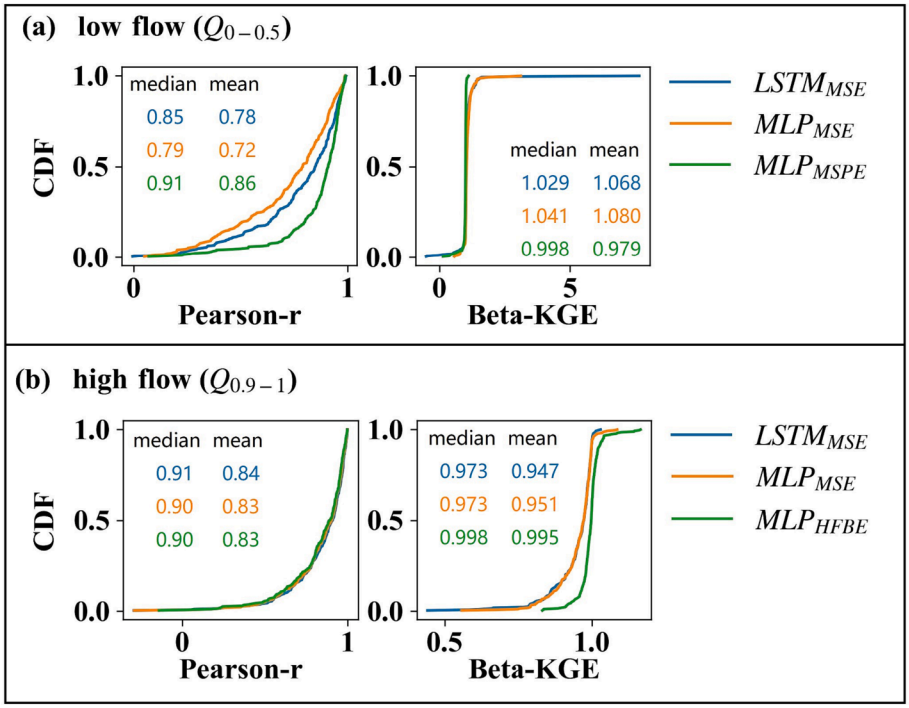


Fig. 10. The cumulative density function (CDF) of the performance metrics for the LSTM and the MLP models trained with MSE, HFBE, and MSPE in (a) low flows ($Q_{0\text{-}0.5}$) and (b) high flows ($Q_{0.9\text{-}1}$). The medians and means of the metrics for $LSTM_{MSE}$ and MLP_{MSE} are displayed in blue and orange, respectively, while the medians and means of the metrics for both MLP_{MSPE} and MLP_{HFBE} are displayed in green. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

prevent floods and reduce disaster.

An important point in building an operational model for a given project is to embody our goal in the objective function. In fact, the design of an application-orientated objective function can be very flexible. MSPE in our study is a transformed version of MSPE that is inherently sensitive to low-flow errors. As for the HFBE used in this study, it is designed by incorporating a penalty item related to the high flow characteristics into the metric MSE. Likewise, we can design an objective function by combining a commonly used metric (e.g. MSE, RMSE or NSE) wtih a hydrological signature (e.g. rising limb density or baseflow index) depending on the project goal. In this way, we can flexibly establish an objective function that is explicitly associated with our goal for the forecast model.

4.4. Apply ensemble method in hydrological modeling

There is possibly more than one goal in a single project. One may expect to achieve multiple goals in the model by using unvarying parameters. However, as stated by Madsen (2000), the streamflow has various characteristics, posing a challenge for hydrological models to sufficiently simulate all aspects with the same set of model parameters. Therefore, utilizing multiple models with different parameter sets facilitates the attainment of multiple goals. In this study, we propose an objective function-based ensemble model (OEM), in which there are three base models trained with different objective functions concerning the low, medium, and high flows, respectively, and integrated by a classification model for the final prediction. Our results show that the OEM can take advantage of each base model and thus outperforms individual base model and the benchmark model, suggesting that OEM plays an efficient role in balancing multiple objectives. Furthermore, it is imperative to highlight that our proposed methodology efficaciously surmounts the challenge associated with the prerequisite of foreknowledge regarding forthcoming streamflow levels, a limitation in the techniques established by Duan et al. (2007) and Sahraei et al. (2020). Consequently, our approach stands as a suitable choice for seamless integration into pragmatic streamflow forecasting endeavors.

LSTM is applied to establish the base models for the OEM in this study. In fact, the OEM can be a flexible framework, where other machine learning methods as well as process-based models can also be used as base models. In the model calibration/training, the optimization goals can be varied as well. This includes improving model performance in simulating the high flow days or the peak flow timing. In addition to the base model, the integrator in OEM can also be flexible. For example, expert knowledge containing the hydrologists' experience can be used to determine how base models should be integrated for the final predictions.

Ensemble methods are not new in hydrological modeling, and a commonly used ensemble method is to establish multiple models with various random seeds and then compute an average of the predictions from these models (e.g. Kratzert et al. (2018) and Kratzert et al. (2021)). On one hand, this approach can generate stable predictions by averaging the outputs from all models. On the other hand, this approach may become mediocre, because the prediction from the good models is accounted for equally as those from other models. Worse still, models trained with the same objective function are highly homogenous and have similar performance (either all good or all bad) under the same conditions. For example, even though a number of LSTM models are trained with the objective function MSE, none of the $LSTM_{MSE}$ models can obtain the same level of good performance as the $LSTM_{MSPE}$ in low-flow forecasting (not shown). Different from the commonly used ensemble method, the base models in our OEM are established through different objective functions, meaning that they are heterogenous and skilled in capturing different aspects of streamflow. On this basis, an integrator integrates them by leveraging their individual strengths, thus achieving a multifaceted prediction. In addition to incorporating a diverse array of distinct objective functions, the inclusion of multiple hydrologic models characterized by disparate architectural configurations stands as a salient avenue for cultivating base models with heterogeneity, as done in the studies by Duan et al. (2007), Sahraei et al. (2020), and Chlumsky et al. (2021).

4.5. Limitation of the OEM

It is crucial to note that the recommended approach is mainly designed for accurate predictions rather than facilitating an in-depth understanding of hydrological processes. While the OEM significantly improves predictive capabilities in streamflow modeling, it poses a challenge in terms of interpretability, introducing complexity to the 'black box' nature of machine learning models. Unlike hydrological models, which employs specific structural equations to represent distinct processes, the OEM's machine learning components lack the same level of clarity in physical interpretation. Future research could explore methods to extract hydrological insights from individual ensemble member models and the classification model within the OEM, potentially addressing interpretability challenges.

5. Conclusions

The objective function plays a fundamental role in the model parameter optimization, and can heavily influence the model performance in hydrological modeling. In this study, we design an objective function, HFBE, to alleviate the high flow underestimation that exists in the models trained with MSE. We also design the metric MSPE which is inherently sensitive to low-flow errors. We establish different LSTM models trained with HFBE, MSPE, and MSE, respectively, and examine the strengths and weaknesses of these streamflow forecast models with multiple evaluation metrics. Furthermore, we propose the objective function-based ensemble model (OEM) framework that integrates the models trained with different objective functions, so as to take advantage of the trained models focusing on different aspects of streamflow and thus achieve better overall performance.

Our results in 273 catchments over the USA show that the models trained with MSE do not perform well in forecasting the low flows. The MSE-trained models also suffer from underestimation in high flows. The models trained with HFBE can alleviate this underestimation in high flows and thus perform remarkably better in resembling high-flow characteristics. It is also found that the models trained with MSPE outperform the other two models in low flow forecast, no matter what algorithm is used for the model establishment. However, the MSPE-trained models perform poorly in predicting high flows. By incorporating the models trained by HFBE, MSE, and MSPE, respectively, our proposed OEM performs well in the forecast of both high flows and low flows, and realistically captures the mean and the variability of the observational streamflow. In addition, the OEM can still play an effective role under four different scenarios with dry, wet, homogenous, and heterogenous hydrometeorological conditions. More importantly, this framework readily lends itself to direct applicability within the realm of authentic streamflow forecasting operations, facilitating a seamless transition from theoretical constructs to practical utility.

This study highlights that the objective function can play a more important role than the algorithm for model establishment, and it is helpful to apply application-orientated objective functions for given projects in the streamflow forecast. This study also demonstrates the potential of ensemble learning methods for balancing multiple objectives in hydrological modeling.

CRediT authorship contribution statement

Yongen Lin: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Dagang Wang:** Conceptualization, Funding Acquisition, Writing - Review & Editing. **Jinxin Zhu:** Writing - Review & Editing. **Wei Sun:** Methodology, Writing - Review & Editing.

Chaopeng Shen: Writing - Review & Editing. **Wei Shangguan:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Open Research

The data used in this study are available at https://ldas.gsfc.nasa.gov/nldas/ and https://www.bafg.de/GRDC/EN/Home/homepage_node.html. The model code and result analysis code are also publicly available via GitHub repository (https://github.com/LinyenC/p).

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